

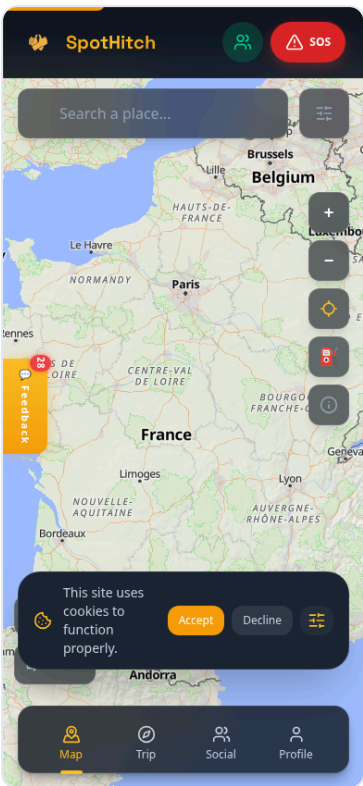
SpotHitch — App Screenshots & Technical Architecture

Attachment for NGI Zero Commons Fund Application — March 2026

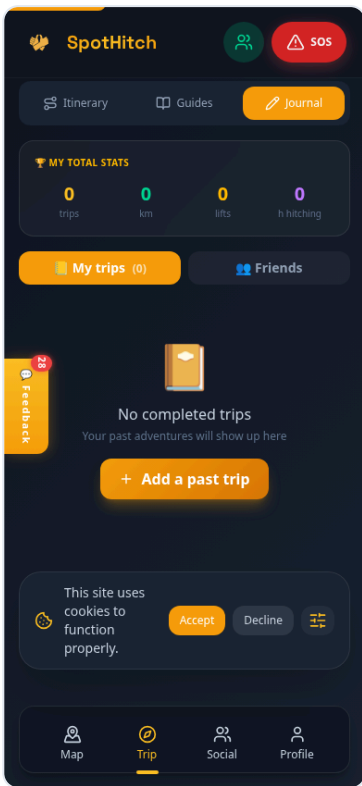
1. Current App (Alpha Version)

SpotHitch is a fully functional Progressive Web App available at **spothitch.com**. It works on mobile (iOS/Android) and desktop browsers. Here are screenshots of the current production version:

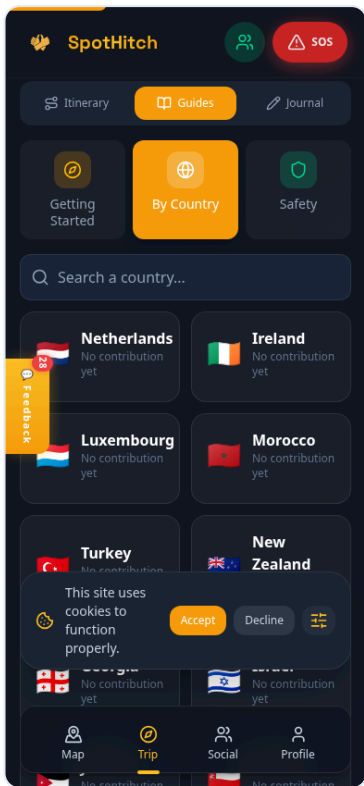
Mobile (390 x 844px)



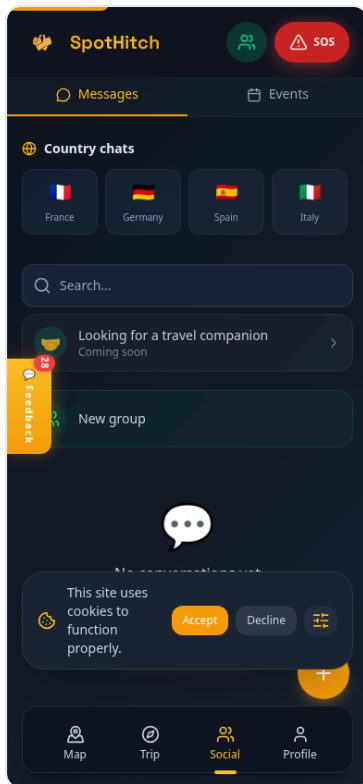
Map with spots



Trip journal

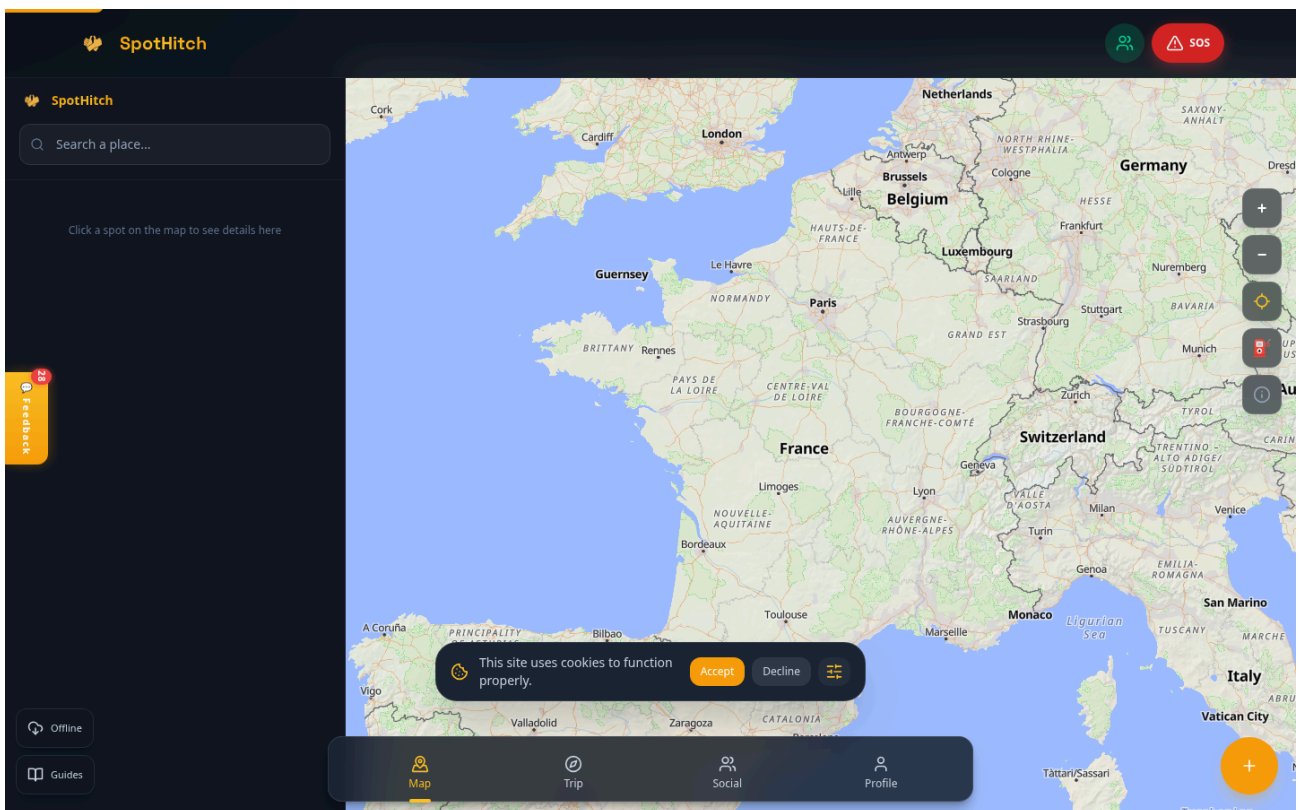


Country guides (96 countries)

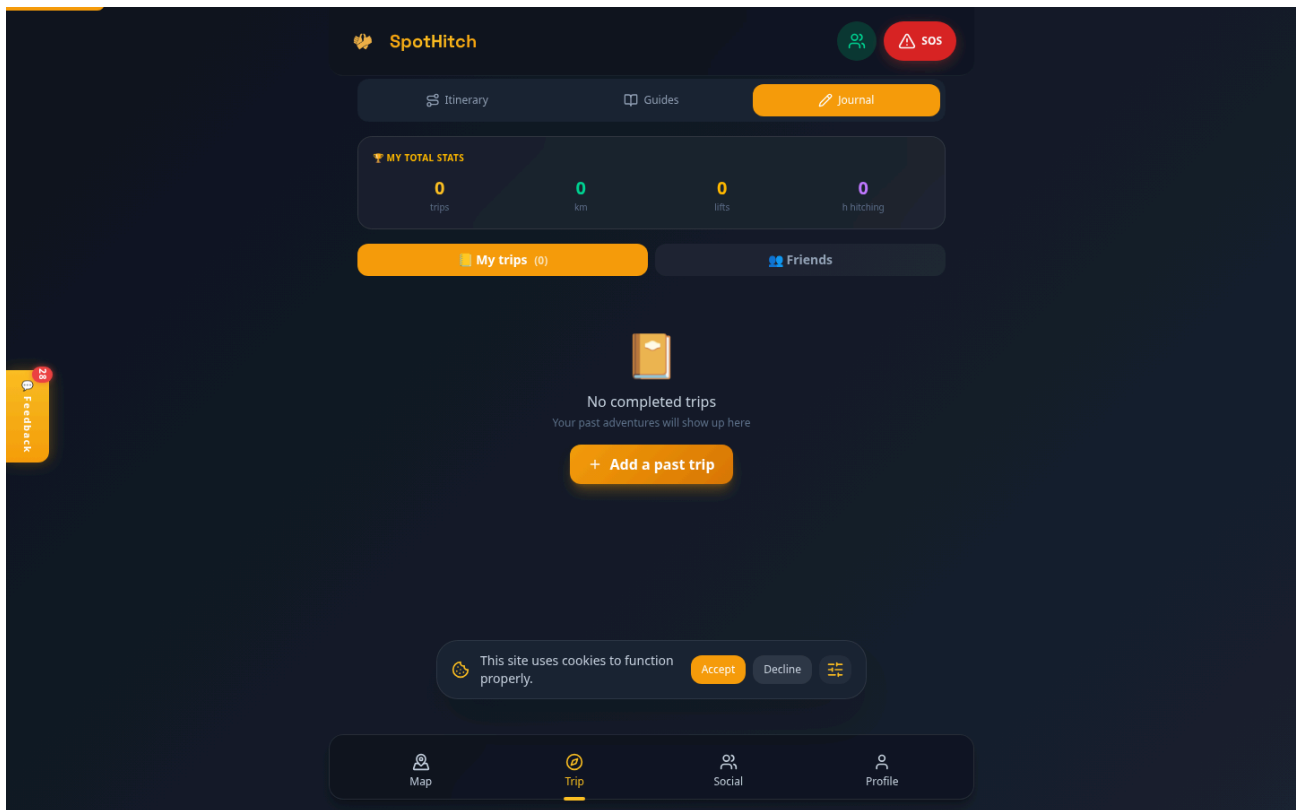


Social / messaging

Desktop (1440 x 900px) — with split-view map



Desktop: side panel + map (Airbnb-style split view)



Desktop: centered content with responsive layout

2. Existing Features (already built)

Map — Community spots with ratings, photos, clustering

96 Countries — Guides with laws, safety, budget, culture (4 languages)

Social — DMs, group chats, country chats, friend system

Trip Journal — Log trips, stats, itineraries

Gas Stations — Toggle overlay on map (Overpass API)

Offline — Full offline mode, sync when back online

Gamification — Points, badges, leaderboard

Privacy — GDPR compliant, cookie consent, data export/delete

4 Languages — FR, EN, ES, DE with dynamic loading

PWA — Installable, works on all devices

3. Guardian Mode — Technical Flow

Guardian Mode lets a traveler share their position in real-time with trusted contacts ("guardians"). If the traveler doesn't check in within a chosen interval, guardians are automatically alerted.

Normal Flow

- 1 **Traveler activates Guardian Mode** — selects guardians (phone contacts or SpotHitch users), sets check-in interval (30min, 1h, 2h, custom)
↓
- 2 **Position tracking begins** — GPS via Geolocation API, adaptive frequency based on movement and battery. Positions encrypted end-to-end before sending to server.
↓
- 3 **Check-in timer** — app prompts traveler to confirm they're safe. Push notification + vibration + sound.
↓
- 4 **Traveler responds "I'm OK"** — timer resets, guardians see updated position on their map. Loop continues.

Alert Flow (traveler doesn't respond)

- 5 **Grace period** (configurable, default 5 min) — repeated alerts to traveler. Distinguishes "no network" from "no response."
↓
- 6 **Automatic alert to guardians** — push notification + automatic SMS (Twilio) + email with last known position, battery level, and trajectory.
↓
- 7 **Community alert** (opt-in) — nearby SpotHitch users receive a notification that someone may need help, with approximate area (not exact position).

4. SOS Mode — Technical Flow

SOS Mode is a set of emergency tools designed for situations where the traveler feels unsafe. All tools are designed to work discreetly.

Emergency Activation

- 1 **One-tap SOS button** — large, easy to find even in stress. Can also be triggered by shaking the phone (accelerometer).
↓
- 2 **Silent alarm** — no visible change on screen. The phone appears normal. Alert sent silently to guardians with position.
↓

3 **Fake incoming call** — realistic phone call UI with configurable caller name. Gives the traveler an excuse to leave a situation ("Sorry, I have to take this").



4 **Audio/video recording** — discreet recording starts automatically. Stored locally, encrypted, can be used as evidence.



5 **Multi-channel alert** — SMS (Twilio) + push notification + email to guardians. Community alert to nearby users (opt-in).

Offline resilience: If there's no internet connection, all data (position, recording, alert) is stored locally via IndexedDB and transmitted automatically via Background Sync as soon as the network returns.

5. Technical Stack

Frontend — Vite 5, vanilla JS ES Modules, Tailwind CSS 4

Map — MapLibre GL JS + OpenFreeMap (open source)

Backend — Firebase (Auth, Firestore, Cloud Messaging)

SMS — Twilio (for guardian alerts)

Encryption — Web Crypto API for E2E encryption

Offline — Service Worker + IndexedDB + Background Sync

Testing — Vitest (87 tests) + Playwright E2E

CI/CD — GitHub Actions + Cloudflare Pages